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Towards meeting futures' educational needs: a case study of CERN IdeaSquare planet

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ABSTRACT

With the future being increasingly perceived as uncertain, education must develop skills to foster sustainability and responsibility in unpredictable circumstances. This study explores how CERN IdeaSquare Planet (I2P), an innovative educational program at the intersection of innovation and transformative futures education, works towards meeting this need. I2P teaches key skills such as transformative thinking, openness to alternatives, and comfort with uncertainty, while also addressing the largely unexplored impact of futures exercises on present behavior. For explorative study of the impacts of I2P on these skills, we study three pilot program runs. We employ both thematic analysis of reflection text data from participants and facilitators, and ethnographic participant observation as methodologies. Findings suggest that I2P develops essential futures skills such as openness to alternative futures, managing uncertainty and complexity, and self-organized transdisciplinary teamwork. However, while openness to alternatives increased, participants ultimately gravitated towards adopting presently dominant future narratives like techno-optimism, limiting the consideration of fully transformative futures imaginaries. This study provides practical insights for educators interested in implementing similar methodologies for transformative innovation education, raises new questions for further research in futures education, and offers novel insights on previously unexplored futures imaginaries' impacts on present problem solving.

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1. Introduction

Both academia and industry are trying to adapt to perceived increasing uncertainty of the future (Lehtonen et al., 2022). One common way of adaptation is through futures education, in which futures thinking and skills are brought into classrooms to create future citizens and professionals comfortable with navigating environmental contexts of uncertainty whilst fostering both responsibility and sustainability of societies and nature (e.g. Laherto & Rasa, 2022; Levrini et al., 2021; McWilliam et al., 2008; Page, 2002; Rasa et al., 2022). Futures education efforts have been embedded into various disciplines and levels: Primary school education has connected futures thinking to sustainability education and to local community practices (Spellman et al., 2021). On a high-school level, science education has adopted futures thinking practices with positive results (Rasa et al., 2022), and in universities futures thinking methods have been adopted across disciplines, including engineering education (e.g. Henderson et al., 2023; Lande & Leifer, 2010; Sumper et al., 2023), design education (e.g. Evans & Sommerville, 2006; Lehtonen et al., 2022), and education policy in general (e.g. Istance & Theisens, 2013). Futures education is also a global effort (Barendregt et al., 2024), with different regions of the world focusing on different aspects of futures. A good overview of futures educational efforts across different global contexts is

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offered by a 2018 UNESCO report (Miller, 2018). This is unsurprising as future-related uncertainties such as global warming are, surely global, but also unequally impactful across the world.

Futures education stems from the futures field, which is a vibrant and developing interdisciplinary field with roots in—for example—psychology, sociology and anthropology (Poli, 2014) with a wide variety of tools and methods. An eclectic history means differences in strands of research. Despite, or perhaps due to, these interdisciplinary roots that can capture possible oversights or simplifications of a single field, there persist multiple open questions, concerns, and points of development to consider in futures education. Out of these, we focus on three. First, fostering openness to alternatives is important for futures education. Since desire is subject to a plethora of variables, one desirable future can not be desirable for all, especially since imagination surrounding futures stems from our own varied lectures of the present. A closedness of futures to certain, naturalized futures narratives, would limit the capacity of groups outside them to grasp their potential of action for desirable futures (Dolan & Rajak, 2018; Wathelet, 2020).

In other words, closed futures is the opposite of openness to alternative futures in how people can imagine the future. Namely, closed futures imaginaries are limited in what is seen as possible to change and tied to largely current narrative trajectories. Hence, futures imaginaires with such lack of plurality make it difficult for people to even start imagining, let alone build, futures worth working towards, which may inhibit sustainability transformations (Köves et al., 2024) and emancipatory efforts, with agency in relation to the futures' being already unequally distributed due to economic inequalities (Appadurai, 2013b; Højlund et al., 2011; Wathelet, 2020). Since the 1990's the "closed future" tends towards the neoliberal view dominating foresight activities with its associated "there is no alternative" narrative, with the rest of the field becoming fragmented (Son, 2015). This narrative, characterized by anxiety-alleviating uncritical acceptance of taken-for-granted "reality", is considered an issue in many future studies, since it can be seen as colonizing the future rather than considering transformative futures (e.g. Bergmann, 1992; Feola et al., 2021; Hutchinson, 2005; Levriani et al., 2021; Liveley & Wardrop, 2020). Young people are having troubles imagining futures outside of the dominant techno-optimistic and eco-modernization narratives, but are not aware of alternatives (Köves et al., 2024). Both techno-optimism and eco-modernization largely point to the dominant belief that science and technology development will be able to solve the complex socio-environmental challenges of our present and future without changes required to growth-based economies and affluent Western ways of life (Alexander & Rutherford, 2019; Köves et al., 2024; Parrique et al., 2019). While weaker forms of techno-optimism focused on promoting human agency remain intellectually defensible (Danaher, 2022), the stronger forms of these narratives may hinder efforts to solve the challenges via reductionist ignorance of the systematic interconnections of these complex problems (Köves et al., 2024) or derailing getting real answers by focusing on maintaining superstitious beliefs about technological potential to maintain current ways of living (Parrique et al., 2019; Wilson, 2017). Therefore, the goal of letting emerge a plurality of future scenarios—openness to alternatives—is key to moving beyond the certainty of fixed narratives towards transformative, emancipatory futures and defying the dominant narratives of the *status quo*. While advocating for a plurality of future representation might be perceived as taxonomist or a simple descriptive work, there is already a this literature showing the negative impact on forcing one future. This is especially important given the fact that future imagination goes hand-in-hand with building a better future. For futures techniques to become more radical, there is the necessity to at least question deeply which kind of future is desirable and who ultimately benefits from dissemination of specific future narratives.

Second, richness of futures educational methods is currently limited: Frameworks and timeframes are largely limited to mostly scenario analysis and roadmaps within timeframes less than 50 years from now (Ketonen-Oksi & Vigren, 2024). Recent futures discourse has called for more methods in favor of difficult—potentially even perceived as threatening (Bentz et al., 2022)—yet necessary changes and transformation to unsustainable powers and structures (e.g. Blythe et al., 2018; Feola et al., 2021; Jasanoff, 2015b; Vogel & O'Brien, 2022). In especially applied anthropology, there is a wish to go beyond the short-term intervention, tackling more radical problems and focusing on long-term solutions when studying futures. For instance, an account made by an anonymous collective based in Notre-Dame-des-Landes, a french rural area occupied by activists fighting against a destructive airport project, is an

example of future narratives that want to emerge as alternative to the ones proposed by institutions or companies. The collective points out the fact that, too often, the same people that are the culprit of the current state of uncertainty of the world will not solve it (H-, 2023). In their own projection workshops, they do not only explore desirable futures in an anti-capitalistic way, but they also strongly reclaim the right to imagine their future, free from the influences of other productions coming from the upper stratus of society. Sociology also foregrounds the source of futures narratives: The “*expectation[s] about the state of things to come at some future time*” are what Bell and Wau (1971) defined as “images of the future,” images that are produced by the elite of the society, adds Tutton (2017). In this vein, Ketonen-Oksi and Vigren (2024) call for “tools and methods that dismantle existing worldviews, values, aims, and perspectives.” Thus, we need methodologies that go beyond technology development on to addressing the underlying societal structures. For example, causal layered analysis and critical scenarios (e.g. Boonmavichit & Boossabong, 2022; Cairns et al., 2010; Son, 2015) are considered methods that may be better at accomplishing true transformative evolution beyond the *status quo* power structures (Bentz et al., 2022; Dlouhá et al., 2019; Vogel & O’Brien, 2022) that emphasise individuality over collective effort. Furthermore, an immersive and an experiential component in futures methods can contribute towards desired outcomes (García & Gaziulusoy, 2021) such as agency and capabilities to manage uncertainty. To meet established futures’ needs, such methodologies ought to be taken up and developed also in education.

Third, like in futures studies in general (Ketonen-Oksi & Vigren, 2024), futures education lacks impact assessment beyond vague “positive impacts” around sense of agency and other internal states (e.g. Ahvenharju et al., 2018; Laherto et al., 2023; Laherto & Rasa, 2022; Levrini et al., 2021; Rasa et al., 2022). Since the first empirical measurement systems were established around these concepts (e.g. Ahvenharju et al., 2018; Hsu, 2020), it is understandable that most work until now leverages these tools. In addition, often future employability is highlighted as a positive, taken for granted, impact (Geertshuis et al., 2024; Mlcek & Bell, 2018; Scott, 2019). Indeed, it is sometimes assumed that it will be specifically the workforce making transformative change for the future (e.g. Kong et al., 2025). This focus on internal, individual change within the framework of neoliberal competitive aspects like “employability” over social and behavioral impacts may explain why transformative ideas born from futures thinking may be left to implement by “someone, sometime” (Wolff et al., 2022). Rarely, if ever, has research focused on how futures thinking skills can aid in solving problems and generating transformations in the now—not only hypothetical ones in the future.

This article explores the first case study of an experimental educational futures’ methodology intended to meet the three identified futures educational needs of openness to alternatives, methodological richness for transformativeness, and impact on the present moment problem solving: CERN IdeaSquare Planet (I2P). While it is acknowledged that transdisciplinary and participatory course designs around critical and systems thinking are key to transformative futures’ educational goals (Kiss et al., 2021; Köves et al., 2024; Lehtonen et al., 2018), I2P has been developed by the IdeaSquare Team at CERN (CERN, n.d) for transformative innovation to go beyond incorporating certain, limited elements towards a holistic educational approach incorporating both qualitative and quantitative critical, systems, and futures thinking in the context of self-organized transdisciplinary teams working on uncertain challenges both in futures and present. As such, it is uniquely set up to tackle challenge-based education around wicked futures-related problems like, for instance, sustainability (Lehtonen et al., 2018). To achieve this I2P employs a variety of futures education methods in a customizable story narrative for a group of students who take on a specific narrative challenge related to the travel to and settling on an exoplanet in the far future. The narrative structure and components can be varied in terms of focus on technology, policy, degree of science fiction and realism, to name some variables. Experiential immersion along with a toolbox of futures thinking methods is implemented in the task to accomplish this mission. The key part is the second half of the project course, in which the learned futures skills are applied to solving a real world problem in the present. As such, I2P contributes towards the identified need for futures education having an impact already in the present. The next section describes I2P methodology in further detail in how it supports futures education’s needs being met. Specifically, by analysing reported learnings and feedback from three pilot runs of I2P, we study how I2P can meet the three identified needs for futures education described above.

We find that students become open to alternatives through exposure to the different opinions of their diverse team members and the uncertain nature of the task given along with the encouragement to think freely. In addition, many students find that they become more comfortable with uncertainty and these learnings were also applied in a task unrelated to the futures narrative, implying a degree transference of learned futures skills to present problem solving. Some noted issues that remain to be improved upon in the I2P methodology are limited capability to counter techno-optimism and deterministic thinking. In addition, prototyping and hands-on activities were found useful to ground abstract discussions. These findings provide tentative yet valuable directions for developing futures education methodologies. In short, I2P met the desired futures capabilities of dealing with uncertainty, openness to alternatives, and transferring skills to present problem-solving. However, there is room for future improvement in the used toolbox's capability to get students to commit to truly transformative and radical futures.

2. Theoretical background

This section begins with a brief theoretical background of futures studies' interdisciplinary roots. After this, the section introduces the core principles of the educational methodology of I2P in reference to these roots. However, it is to be noted that all I2P programs are tailorable and, hence, this section can only address the core elements present rather than fully cover the program.

2.1. Futures' interdisciplinary lenses

Currently, futures is widely interdisciplinary, but has extensive history across social sciences. Namely, psychology, sociology, and anthropology have developed their own distinct streams of literature around futures with similarities and clear differences in frameworks and foci across these fields. For instance, a central endeavor in psychology has been to try to understand the causes of human action and, historically, the focus in psychological theory and practice has looked at human action as a consequence of the past (Seligman et al., 2013). However, more recently, with explaining the actions of the present by how the future is perceived, called *prospection*, or *future-directed thinking* (Gilbert & Wilson, 2007; Szpunar et al., 2016), there has been a shift from looking at past experiences to researching the perception of the future when it comes to explaining human behavior. Anthropology's interest for the future is quite recent, due to the discipline's tradition of studying social dynamics in present-time via *ethnography*. Now, however, anthropology of the future's literature is blooming with future-related issues being published in major anthropological journals, such as the "Speculative Anthropology" issue in *Cultural Anthropology* (Anderson et al., 2018) or the special double-issue "Futurofolies" published in *Terrain* (Grimaud & Wacquez, 2023). A review of literature by Wathelet (2020) shows an interest in not only describing different ways of conceiving futures—meaning that future is not perceived the same way in each culture—or different techniques—scenarios, projection, divination, imagination—but also analyzing the power of futures to shape the present and to let emerge a desirable world (Wathelet, 2020). Futures in sociology can be traced back to 1971 when Bell and Wau (1971) published "Sociology of the Future: Theory, Cases and Annotated Bibliography," as a reaction to their dissatisfaction with traditional sociological orientations that primarily focused on the present and past, and as a desire to redefine the role of social scientists as advisors to policymakers. The field never really took off, however, as Bell noted two decades later (Bell, 1996). In this sense, it can be said that sociology of the future has never really left its status as an emerging field. It has nevertheless developed through various theoretical and methodological approaches aiming to analyze futures consciousness and its implications for social life. Selin (2008), for instance, describes the field's dynamic as "*drawing from a mix between Science and Technology Studies and the practice of foresight*" (p. 1878). She emphasizes the performative role of expectations in innovation and the importance of studying how communities use anticipatory knowledge, as these understandings could provide "*means of taking responsibility for what is to come*" (p. 1878). She posits that society not only changes with the usage of technology, but that it also produces meaning while the technologies develop. With the acceleration of processes of discovery and invention, our capability of making sense of the future is challenged. This is not insignificant, since, as (Selin, 2008)

illustrates with nanotechnology, the rhetorical construction of the future influences which technologies are brought to existence. Overall, in comparison, psychology emphasizes futures imaginaries' impact on human behavior, anthropology emphasizes transformativeness for social justice, and sociology brings forward the role of technology in shaping society in futures imaginaries. These particularities of the fields also inform the structure of I2P.

2.2. IdeaSquare planet: concept and methodological outline

IdeaSquare Planet (I2P) is an educational initiative developed by the team at CERN's IdeaSquare facility. IdeaSquare is designed to foster early-stage innovation through interdisciplinary collaboration (CERN, n.d). IdeaSquare brings together students, scientists, and organizations from various fields to work at the "Fuzzy Front End" of the research, development, and innovation (R&D&I) process, focusing on the early phases of idea generation and experimentation. By aligning its efforts with the United Nations' Sustainable Development Goals (SDGs), IdeaSquare connects CERN's fundamental science to addressing global challenges (United Nations, n.d). Built on the foundation of futures thinking methods, I2P equips master's level students for tackling complex, or "wicked," problems (Rutting et al., 2024) with essential skills and mindsets. Wicked problems, by definition, are difficult or even impossible to solve due to their incomplete, contradictory, and constantly evolving nature, where key requirements are often uncertain (Lönngren & van Poeck, 2021). The term "wicked" signifies the resistance these problems pose to resolution, with global sustainable development falling into this category (Kerekes, 2023). In response, I2P focuses on teaching futures skills such as systems thinking, both qualitative and quantitative reasoning, strategic planning, and decision-making under constraints, therefore preparing students to navigate and address such multifaceted challenges in transdisciplinary teams (Hull et al., 2020).

Indeed, I2P equips students with a versatile toolbox of skills for identifying and deconstructing the various dimensions of complex problems. It consists of essential modes of thinking and analysis as mentioned previously, such as systems thinking, critical thinking, hypothesis formulation, and both quantitative and qualitative impact reasoning. By employing these tools, students are able to break down intricate, multifaceted issues into their core components. It is only after this thorough deconstruction that they can gain a comprehensive understanding of the "wicked" problem at hand and such problems in general, enabling them to attempt a novel and well-informed reconstruction aimed at developing viable solutions. In this context, I2P frames sustainability as a prime example of such complex challenges—one that requires a dual approach of deconstruction (critique through quantitative and qualitative analysis) and reconstruction (action and hope grounded in these same analyses) (Satterwhite et al., 2020). This framing allows also the possibility for the student teams to explore sustainable futures through of societal value change (Milbrath, 1984). To achieve this, I2P guides students through a two-stage journey in a workshop lasting at least one week with a central, *status quo* challenging, overarching question posed to them: *If humanity had the chance to start over, how could we avoid repeating the same mistakes that led to Earth's present unsustainable state?*

In the first stage of the program, students are immersed in a narrative where humans seek to explore an imaginary exoplanet for a variety of reasons depending on the storyline, such as having to abandon Earth due to a crisis or for a science mission. Humanity is now challenged with inhabiting an exoplanet—"Planet Y." Within this framework, teachers—meaning the teaching staff from respective universities—and facilitators—meaning the educational staff at CERN IdeaSquare—collaboratively design a narrative that presents students with specific challenges based on one or more of the following themes: who and what to take on and how to reach Planet Y, organizing travel, establishing a first settlement, and fulfilling the settlement's mission. Each theme represents a complex, "wicked" problem. To ensure the challenge is achievable within the program's timeframe, educators carefully structure the narrative and tasks, maintaining much of the challenge's complexity but ensuring with some set boundary conditions that it can be tackled within the given workshop period. These challenges may include, selecting the profiles of the first settlers, managing limited water and food resources in the first settlement, setting up a basic healthcare unit, and/or establishing foundational principles to guide decision-making. This first stage concludes with students presenting their solutions to the crafted challenges. These presentations must include a clear formulation and justification of the assumptions made, supported by

both quantitative and qualitative reasoning, as well as an analysis of the outcomes that resulted from those assumptions. The hypothetical scenario allows students to challenge many preconceived boundaries and biases that typically surround complex real-world problems on Earth. The speculative setting frees students from existing assumptions, encouraging them to think more creatively and to explore novel, potentially transformative approaches to the challenges at hand.

“There is no Planet B” is a phrase often heard in climate action, meaning that we must take care of Earth because there is no alternative. This sentiment is fully acknowledged by CERN IdeaSquare. The terminology of “Planet B” is avoided in the detailed narratives given for students and usually a more specific exoplanet is given, e.g. Trappist planets or specifically “Planet Y.” The “Y” (why) also reflects the need for students to question the existing systems on Earth, and to reconsider systems instead of copying them. Planet Y serves as a blank canvas to paint novel systems—within the limits of given boundary conditions—for participants to come back to and as an inspiration to transform and disrupt the systems of today. Such sociotechnical imaginaries are conceptualised by Jasanoff (2015a) as a co-product of science and technology as central institutions, constructing the future to direct our present. She emphasizes that in this fabrication of the future, science fiction is a repository of sociotechnical imaginaries, stressing that fiction has the power to shape the future. Fiction and imagination must therefore be taken seriously since they show that other worlds are possible. Indeed, thinking of a different planet and a society from the start functions as a tool of thought to escape the “there is no alternative” narrative, in which students may disregard possibilities to organize society for cultural and normative current constraints. Moreover, using a narrative that students get to shape and make up what the future could look like in diverse teams can encourage them to see the future not as something that “exists,” but is made, while remaining reflective over such ontoepistemological questions (Ketonen-Oksi & Vigren, 2024). To give a concrete example, it is different to consider setting up industrial agriculture on another planet than to imagine overhauling agricultural practices on planet Earth. To challenge assumptions and become open to alternatives, the future and conventional frameworks for thought need to be undefined (Inayatullah, 1998; Sardar & Sweeney, 2016) and using starting anew somewhere new is an effective and creative way to shed these presumptions—as long as students are not left on a “Planet B” but brought “back to Earth” to apply their learnings.

In I2P, this first part functions as an open exploration of futures narratives, but the emphasis is always heavily on the second half, in which students “return to planet Earth,” shifting from the speculative scenarios of Planet Y to a grounded, real-world context. This reflects the intent in futures psychology for developed perceptions of future (the first I2P phase) to impact behavior in the present. Indeed, in the second phase of I2P, students are presented with a tangible, complex short-term future scenario based on real-world trends or scientific estimations that requires them to apply the futures skills they acquired during the first phase of the program and to translate their gained skills into practical, actionable solutions. These cases are designed to reflect the wickedness of the most pressing global issues, such as managing the aftermath of environmental disasters, addressing the humanitarian needs of immigrants and refugees in overcrowded camps, or developing innovative strategies to drastically reduce urban air pollution from vehicles. In tackling these scenarios, students must not only craft thoughtful and well-justified action plans but also back up their recommendations with both qualitative reasoning and quantitative analysis. Each team must demonstrate a deep understanding of the problem at hand by covering a wide range of variables, leading to many potential entry points and both local and global opportunities for intervention. This second phase emphasizes the practical feasibility of their proposals, asking students to adapt or eventually completely challenge the *status quo* while implementing their transformative solutions within the limitations and complexities of reality. This way of approaching wicked problems is not context-dependent and, hence, is also applicable to other challenges students will face in the future.

2.3. Pedagogical cornerstones of IdeaSquare planet

A comprehensive description and analysis of the pedagogical aspects of the I2P program is beyond the scope of this article. However, it is worth highlighting several key elements. I2P is a tailorable program

to fit constraints and wishes from participants. Yet, I2P always utilises specific methods for futures education. These are explained below along with their rationale for inclusion in the program.

2.3.1. Transdisciplinarity

In the spirit of CERN, I2P is fundamentally multidisciplinary and increasingly transdisciplinary. Transdisciplinarity means that students do not simply contribute their own area of specialisation to teams, but rather new avenues of activity emerge that require overcoming disciplinary lines altogether to form language and understanding. McWilliam et al. (2008) take a note from a lecture given by Charlie Leadbeater in 2004 and emphasize that transdisciplinarity can be seen as useful ignorance towards different realms of knowledge that can take us forward rather than being stuck in the comfort of what we know. In addition, transdisciplinarity can foster an environment of equity in the context of a “blank slate” planet regarding different futures imaginaries, providing an opportunity for openness to alternatives based on a variety of imaginaries to be considered. While transdisciplinarity is key to transformative futures thinking (Ketonen-Oksi & Vigren, 2024), it requires specific facilitation to elicit wanted learning outcomes. For instance, for academics, transdisciplinarity may be anxiety-inducing and uncomfortable (McWilliam et al., 2008). Indeed, facilitation requires sensitivity to social dynamics and potential issues to success, such as the aforementioned anxiety, resistance to the process, and potential unwanted power dynamics unfolding between marginalized individuals and the explored futures (Finn & Wylie, 2021; Inayatullah, 2006; Ketonen-Oksi & Vigren, 2024; Nabavi, 2022).

For instance, McWilliam et al. (2008) go on to suggest that open spaces can be key for transdisciplinary work's success. Openness of work and space is deeply at the heart of IdeaSquare: A fully transdisciplinary team and an open area to complement the open minds. With IdeaSquare as a central open space in the I2P programs focused on problem-solving around wicked and global challenges and expert facilitation for diverse teams if provided, I2P is set up to minimize any issues pertinent to transdisciplinarity. Trips, such as those to IdeaSquare, facilitate students learning independent questioning, critical thinking and collaboration (Adefila et al., 2023), which are all considered vital for successful futures education.

2.3.2. Narratives and storytelling

Narratives and visions of future are posed as ways for people to become engaged with futures (Elder, 1995, e.g.). Imagined futures are key to how people reason in situations of controversies around social, technical and environmental issues (Chateauraynaud & Debaz, 2017). For instance, Coulter et al. (2019) suggest that climate knowledge may be overlooked if narratives of futures are not engaged with. Hervé and Panissal (2022) suggest that futures education requires questioning paradigms of science and technology and their development to deconstructing normative, current Western thinking that has been based on the former for centuries. Hence, sci-fi narratives become key to exploring, in line with the sociology of futures, how technological aspects and imaginaries are tied with conceptualizations of society. This is especially important, as people's concepts of a good future will inevitably differ: One's utopia can be another's dystopia. I2P necessarily includes sci-fi elements along with transdisciplinarity and thus invites the type of deconstruction invited in the research literature: Narratives combined with diversity are vital to any deconstruction effort, because people's relation to futures differs based on backgrounds, which can bring forward emancipatory points for futures imaginaries. Different social groups are differently positioned to perform their visions of the future (Appadurai, 2013a), which can affect the way different experiences influence futures orientations. Indeed, bringing together different imaginaries can highlight asymmetries and enable addressing them. In I2P diversity aims at reconciling differences to create completely new narratives for futures that everyone can share and find hope in.

However, construction of novel futures narratives and scenarios is not a simple task. Currently narratives and imaginaries of the future focus around either utopias, dystopias, or continuation of business as usual (Bennett et al., 2016). However, extensions of preferred futures now should not be attempts to colonize a future that we ultimately do not know and, as such, we should avoid imposing things into it and rather explore possible futures with an open mind without a normative mindset (Liveley & Wardrop, 2020; Poli, 2017). Something that can aid in surpassing the typical narratives could be exploration of “wild card” events, which can prepare us for that for which there is a lack of experience and to combat

and pinpoint points of ignorance (Nikolova, 2017). I2P incorporates inclusion of wild card events, and other potential modifications to fit needs of each program according to time available or educational interests. These modifications can also be used to re-direct student groups towards more imaginative solutions, to support them in case they get stuck, or to add more complexity if seen as beneficial. Ketonen-Oksi and Vigren (2024) found that futures studies rarely consider timespans beyond 50 years and suggested that this may be due to difficulties of the participants of shifting from considerations of the past to the far future. In addition, Evans and Sommerville (2006) found that the historical path of the current issues is often overlooked altogether in design education. However, such issues can be mitigated with imagery and narrative exercises (Markley & Burchsted, 1997). Hence, the flexible use of sci-fi narratives in I2P is built to facilitate well diversity and range of various futures timeframes and, as such, works towards meeting the established needs of the futures education landscape.

2.3.3. *Self-discovery and uncertainty*

A core principle of I2P is emphasis on self-discovery, which is related to learning to navigate and tolerate uncertainty: Students organically develop key skills such as systems thinking, critical thinking, hypothesis formulation, and impact-driven reasoning, supported by both qualitative and quantitative estimations and evidence. This self-discovery process unfolds as students engage with the challenges presented to them, without being given explicit instructions on what these skills entail or how they have been conceptualized by different scholars, nor are they provided with explicit guidelines on how to communicate or cooperate. They independently establish collaboration and information-sharing dynamics. Students learn these skills “on the go” as they navigate the complexities of the programs’s tasks and develop essential skills through direct participation and exploration, rather than through hierarchical, top-down instruction. It is only at the conclusion of the workshop that a wrap-up session is provided, offering a theoretical framework and context for the skills they have already internalized. This approach encourages experiential learning, where understanding is deepened through practical application, rather than through traditional instruction, intending to make the acquired skills more intuitive and enduring.

Indeed, I2P departs from traditional, formal instruction in favor of a pedagogical approach where the first challenge for educators is “learning to unlearn for relearning” conventional teaching methodologies, processes, and mindsets (Wilson, 1988). In this model, teachers and tutors shift from being authority figures to facilitators of active engagement. Instead of fostering a passive, submissive learning environment, they are tasked with cultivating and encouraging a collaborative dynamic that promotes inquiry, dialogue, and mutual exploration by focusing on acting as sounding boards, providing feedback on students’ hypotheses, estimates, and reasoning, but do not offer explicit solutions and focus on asking questions. Teachers emphasize that there are answers, but no “right” answers and help students deconstruct the complexities and dimensions of the challenge at hand. A related pedagogical principle is the emphasis on seeking a deep understanding before rushing into solutions when dealing with complex, uncertain problems. In these cases, jumping straight to problem-solving can lead to oversimplification, where critical assumptions are taken for granted and key nuances or trade-offs inherent in the problem may be overlooked or misunderstood (King, 1999). By encouraging students to first explore the full scope and intricacies of a challenge, this approach helps prevent misperceptions and ensures that solutions are more thoughtful, well-informed, and capable of addressing the underlying complexities.

2.3.4. *Critical and systems thinking*

Critical and systems thinking are common goals in futures studies, and essential to I2P challenges given to students as missions: Managing limited water and food resources, setting up a basic healthcare unit, and similar tasks taken up by the teams are always interconnected, and students need to collaborate and discover the connections and systemicity of the challenge via self-discovery. Social critical thinking (Larson et al., 2024) in the form of challenging norms, institutions, understandings, as well as peoples’ own thinking are considered aspirational goals in futures studies (Ketonen-Oksi & Vigren, 2024). Indeed, I2P combines both *individual*, meaning the cognitive and affective ability to analyze and evaluate different options for specific outcomes, and *social*, meaning the ability to challenge and question the societal and institutional status quo, critical thinking (Larson et al., 2024). This is an essential part of I2P, but

another way of individual critical thinking (Larson et al., 2024) is also emphasized, namely, grounded empirical calculations.

The goal of this is best illustrated with an example: The classical neoliberal problem-solution to air pollution would be personal filters to sell to individuals to make a profit. However, calculating how much production would be needed for a large city to implement such air filters would reveal that the “technology to solve it” is impossible to scale up or would simply produce too much waste. This requires a new level of critical thinking that is grounded in tangible calculations. Indeed, futures studies tend to lack groundedness as a negative feature of futures imaginaries. For instance, prototyping is proposed as a way to imagine alternative futures, but thus far, nothing has been said about ruling out futures based on such tangibility. This, of course, does and should not limit considerations of “what if” it was possible and eventual learnings from doing so. I2P necessarily includes both, reality checks as a way of critical thinking as well as exploration phases to discover new possibilities. Sometimes constraints from calculations may help think further and more creatively, while maintaining tangibility of the explored futures. Indeed, such reality checks can demystify dominant techno-optimistic narratives and combat tendencies to trust technology to solve the problems of the future while ascribing mystical qualities onto technology (Cook, 2016). I2P, in this methodological mix, combines the intent from anthropology to analyze power structures to shape the present with the sociological focus on how technologies shape the emergence of desirable futures.

2.3.5. Other tools and methods

In addition, immersive methods to thinking like prototyping, embodiment and storytelling foster affective engagement with futures, which can support increased openness to alternative futures (Ketonen-Oksi & Vigren, 2024) and towards desired outcomes (García & Gaziulusoy, 2021) such as agency and capabilities to manage uncertainty. I2P features heavily varied immersive methods and timeframes, with programs always including prototyping and storytelling, but also role-playing, board games, acting, and making videos, are often included in programs, just to give a few examples.

3. Methodology

We employed an explorative methodology to study how I2P contributes towards meeting futures relevant skills of fostering openness to alternatives and uncertainty, methods to critically examine the status quo, and impacts on present behavior. The goal was to discover broadly the impacts of I2P in order to further develop both the program and research on futures education for transformative innovation. An explorative methodology is justified, since I2P is a fully novel educational approach, and as such, no previous research-based knowledge exists. Hence, preliminary exploratory research knowledge is necessary to generate initial emergent insights on what are the impacts related to these goals. The study here covers three separate, week-long I2P pilots at IdeaSquare, during which written feedback was collected from students and voluntary comments from teachers. This was organized as hour-long feedback sessions during the weeks, in which participants and teachers wrote their feedback on unlimited post-its on whiteboards assigned to the topics: about what they liked, what was challenging, what could have been changed, what they learned, and what they take away from the program. These notes were then digitized. This yielded a set of 312 feedback notes from students and teachers. An additional reflection session on similar topics to those in feedback from participants among I2P facilitators yielded 55 feedback/reflection notes on the pilots. During a joint session among facilitators, all these notes were grouped by both source (students, teachers, facilitators) and into three thematic categories: “Openness to alternatives,” “Methodological particularities of I2P,” and “Back to Earth” representing the highlighted research gaps, with “back to Earth,” representing how students related to solving a real-life challenge after the futures exercise part of the program. The coding was done in a shared template by six independent coders, in which the coders individually classified the data into the themes. In this process the coders could see codes assigned by others, and could sign a differential code to a datapoint, but this could not be discussed yet. By doing so, multiple different views were allowed to emerge, mitigating individuals’ subjective biases from the coding. Altogether eight instances of differential codes emerged

regarding either individual datapoints or clusters of similar datapoints, e.g. around teamwork: “How to collaborate in big groups,” “Improved teamwork skills,” “New skills of co-working,” etc.

After all feedback was coded by theme, a validation session was held, in which the six coders reviewed the classifications and marked perceived further inconsistencies for later clarification. All marked differential codes and inconsistencies were discussed to resolve and clarify the coding differences. Finally, an inter-coder agreed-upon coding emerged, in which 60 of the students’ and teachers’ notes were seen as unrelated to any theme, e.g. wishes for better food at the CERN cafeteria, 55 were related to openness to alternatives, 146 were related to the methodological specifics of I2P, and 51 were related to “back to Earth”. From the facilitators’ reflection session eight comments were related to openness to alternatives, 43 to I2P methodological specifics, and four to “back to Earth”. From the aggregate categories, emergent and recurrent subthemes were extracted by the original coders. Namely, the original coders were presented with the task to group similar datapoints together, this time in a shared discussion similar to a focus group methodology, to find recurrent subthemes in the feedback. Multiple recurrent subthemes were found in each category. The results are presented in [section 4](#). In addition, a researcher specialized in ethnography did participant observation for the programs in question. The researcher also did thematic analysis of observations according to the same coding scheme, from which subthemes were extracted by the researcher in a similar manner to, but independently of the feedback thematic analysis. The findings between the observational data is compared with the text based feedback data to find common threads to increase the validity of the done research. All results are presented in the following section.

4. Results

4.1. Impact on openness to alternatives

4.1.1. Students

In the results of I2P impression on students, openness to alternatives was a core theme with students reporting almost exclusively two themes in relation to increased openness to alternatives regarding the futures narrative task posed to them in the program. The first theme in student responses highlighted increased openness to others’ points of views for the future. Examples of comments in this theme were *“I learned how to analyze problems from other people perspective (problem solving in diverse teams),” “I learned to try and listen more of different ideas, even if it sounds insane, because of the task all ideas felt insane,” “nice to try to understand how every person in my team thinks, because, for example, there was an engineer, a designer, and we have like different way of thinking and try to come to explain to each other our ideas and the way we are generating them”, and “Diversity of team members and diversity of opinions.”* While it is not a surprising finding, it was an emergent theme due to the transdisciplinary nature of the program that was not explicitly sought out for.

In the second theme of opening up to alternative futures, students reported taking away from the experience lesser constricted thinking and openness to critically examine ideas they would have considered too eccentric before. Examples of comments in this theme were *“Be less constrictive in my thinking,” “That even seemingly crazy ideas are worth considering,” “The challenge was widely open allowing us to boost our imagination,”* learning to *“Think out of the box but have arguments to justify your ideas,” “Having the freedom to imagine unusual solutions to unusual problems”* and taking away *“New perspectives on my own ethics.”* In addition, some students reported increased openness to ambiguity and uncertainty, e.g. *“I improved my creative thinking skills as well as collaborative skills. I learned that there is no need to be afraid of abstract and imprecise tasks anymore.”*

However, the observational data finds both support and contradictions to this reported increased openness to alternatives. In terms of support, it was found that working together with people of multiple perspectives towards a collaborative future society resulted in teams being open to these shared and compromised visions, despite communications and discussions to create them being notoriously difficult around the alternatives posed from different points of view. Hence, the first theme of openness to different views and perspectives by way of being exposed to the opinions of others finds support. However, regarding the second theme, observational data found only limited openness to alternatives

with narratives of techno-optimism and pessimism about humanity for the future dominating the projects over different potential views. To give an example, a narrative the ethnographer finds representative is that of participants imagining a society in which artificial intelligence would become a god-like figure, a charismatic leader that would unify all humanity, which the participants perceive as being bias-free, unlike humans. When asked about this type of reasoning, the participants may provide arguments around plurality of religions leading to irrationalism, and that only a perfect rational entity, therefore non-human, could help humanity to progress and that this type of scenario was already similar to what we are currently living today, with people asking advice from artificial intelligence assistants, and that those assistants leading humanity better than humanity would lead itself, would be the result of a natural progress of technology. This can be an indication that students feel like they consider openness to many, but in the end tend to fall back on familiar, Western, already dominant narratives after initial exploration.

4.1.2. Teachers and facilitators

Text data from teachers and facilitators echoed the findings of the ethnographic data on the students: They report many points for improvement and noted issues, despite students reporting increased openness to consider alternatives out of their usual pool for ideation. For instance, the level of innovation was considered constrained and limited to “business as usual,” e.g. *“A lot of solutions not so radical; students discarding innovation.”* Suggestions for how to achieve more transformative ideas diverged. Namely, teachers recommended a more structured program, with notes like *“Formulating clearly and concretely what the challenge is helps”* and *“Very specific task worked well as a basis to build on.”* Facilitators of I2P on the other hand saw a problem in teachers pushing for techno-centric solutions to systemic issues. Examples of notes in this vein are *“Teachers making it sound like there is one solution to find [...]”* and *“Some external teachers still push on techno-solutionism.”* Additionally, facilitators suggested that innovativeness of ideas and level of transformative thinking could be bettered with more available time for the students to acquaint themselves with the challenges: *“Students need time to get comfortable.”* The observational data also highlighted that teachers and facilitators felt that the program outputs remain superficial and reproduce very Earth-like conditions in final deliverables despite exploration of alternatives. However, teachers note that students formulate their assumptions well and facilitators noted that it is specifically uncertainty that sparks the initial exploratory conversations towards openness to alternatives in teams.

4.2. Findings on methodological characteristics

4.2.1. Students

Futures studies have posited deconstructing existing worldviews (Ketonen-Oksi & Vigren, 2024), increased agency, and capability to deal with uncertainty as desirable outcomes. This section considers the capability of the I2P methodology and toolkit to produce these results. As the main finding, students highlighted the complexity and uncertainty combined with freedom as the defining experiences and learning to deal with those as a key take-away from I2P. Examples of learnings in this theme were *“Opportunity of transforming a vague inquiry to a more definitive solution,” “Opportunity to manage our time freely,” “To define the details of the problem being solved and take them all into account in the solution,”* and as a take away learning *“How quickly we can adapt.”* However, this was also considered a challenge with multiple comments on the methodology highlighting that *“It is hard to work without structure.”* While challenges may be learning experiences, this feedback from the students did not make a connection between the two and, thus, a conclusion cannot be inferred whether the students felt this as a learning experience or not.

Students reported learning systems thinking due to specifically I2P’s methodological combination of transdisciplinarity and self-organization. Examples of notes of such learnings were *“The problem solving exercise was actually interesting and all teams were integrating their ideas and solutions smoothly,” “How the whole narrative became interconnected quite strongly,” “I liked the way our team managed to find out solutions to solve problems, the way how we divided workload,”* and *“We didn’t gloss over the meta-meaning of this program e.g. it teaches us teamwork.”* This learning did not come without challenge, with

students noting that discussions often waxed and waned and could have been grounded by implementing more of concrete methodologies like prototyping: *"Way too long discussions,"* and what was found challenging was *"knowing when to stop the discussion."* The observational data supports both findings: Students found communication difficult, but also that this challenge could be overcome and transdisciplinary teamwork began to flow better. What became present in the observational data that was not found in the feedback dataset was that communication issues may coincide with issues of inclusivity of women, especially women of colour in the diverse teams, i.e. marginalized team members may find it difficult to get their points of views heard.

4.2.2. Teachers and facilitators

Teachers' notes on methodological points mainly highlight the success of mixing the mission teams at some point, emphasizing the beneficiality of collaboration of diverse people on complex issues. Facilitators agreed with the students that more prototyping and hands-on grounding is needed in the program in the future, and also suggested varied deliverables from the students to enable them to express their process and learnings in a more flexible manner than simply presenting their work in front of varying audiences at the end of the program. In addition, facilitators highlighted that running the program and meeting learning goals is easier when the methodology is planned and co-designed with the teaching team. In the observation data, while some teachers pushed for techno-solutionism, some teachers focused on the methodological process. This finding discovered a more flexible teaching style that remained unreported in the text data. Moreover, it was observed that also both teachers and facilitators found that despite the challenges in communication being large, they could be overcome.

4.3. Impacts on the present moment problem-solving

4.3.1. Students

One key missing component of futures studies is impact assessment on actions and behavior over attitude and employability. In I2P, the learnings from the futures exercise are afterwards applied to solving a present world problem, thus, applying the learnings immediately. This section covers the students' learnings of this part of I2P, implying the futures learnings put into action. Many of the same learnings were mentioned and emphasized in take-aways here. Specifically, openness to alternatives and different ways of seeing problems were highlighted in the second part learnings in addition to confidence in dealing with complex problems in transdisciplinary teams. Examples of notes specifically related to this "coming back to Earth" part were: *"I learned to listen to other people opinions," "I learned different points of views, new tactics to tackle certain issues. I learned that combining people with different backgrounds and cultures open a new perspective and overall increase a quality of the final outcome," "In this problem solving exercise I have learned to think out of the box and understand other people's perspectives," "it's to break down complex problems that cannot be solved by a small team or a single person, and to share the complexity among different, let's say, business teams that work together," "New ways to solve, analyze and work on complex problems as big and small teams,"* and learnings of *"How uncertainty can impact overall performance."*

While a common sentiment among students was that they did not feel a clear connection between the two parts of the program or that the return to Earth was boring, e.g. *"Didn't get the link between task 1 and 2," "That part 1 would be a full week challenge and no part 2," "[Wish] [t]hat there would not be 2nd part," "That second part will be an extension for first part."* The felt separateness of the tasks despite similar learnings suggests that any transferred skills from part one are not due to the connectedness or similarity of the tasks, but rather suggest that the learned skills may apply to a variety of different complex and uncertain tasks. In the observational data, despite students again feeling increased openness to consider alternatives, a common narrative of tech-optimism emerged assuming technology and technology only can provide "neutral" decision-making.

4.3.2. Teachers and facilitators

Facilitators noted the felt lack of connection between the two parts, and suggested this as one potential reason for unmet expectations for transformativeness. Interestingly, teachers left no written notes on the

second part and in the observational data, teachers pushed for techno-solutionism in the second part also, but this time there was no counter focus on the process. In addition, both teachers and facilitators, despite students struggling with uncertainty, felt like students felt accomplished and happy with dealing with uncertainty, but only afterwards.

5. Discussion

Futures education is vital to fostering skills to deal with the perceived increasing uncertainty of the future whilst fostering also responsibility and sustainability of societies and nature (Laherto & Rasa, 2022; Lehtonen et al., 2022; Levrini et al., 2021; Rasa et al., 2022). This paper explored a novel futures education methodology, CERN IdeaSquare Planet, in relation to how it meets specified needs of the futures studies field. We found that the sci-fi narrative of the first part of the program was considered to be more engaging and interesting by the students than the second part of tackling a present problem on Earth. This is expected based on the knowledge that narratives of the future are ways to increase engagement (Elder, 1995). Students largely reported increased openness to alternatives as a result of specifically the self-organization of transdisciplinary teams and the systems and critical thinking tasks of the I2P program. This indicates that I2P was to a degree successful in meeting the goal of futures studies to challenge norms, institutions, understandings and especially the participants' own thinking (Ketonen-Oksi & Vigren, 2024). However, I2P was not as successful in preventing techno-optimism and deterministic thinking and, thus, development ought to be made to challenge the youths' mystical and all-capable conceptualization of technology (Cook, 2016). As it was found that such a push comes from teachers who are not the facilitators of I2P, an intervention to student-teacher interaction could be beneficial to reducing techno-optimism and broaden the openness to other narratives. This was particularly pronounced in the "coming back to Earth" part with teachers emphasizing technical solutions, even if the "Planet Y" part was more open to exploration.

Clearer outcomes were increased capabilities to deal with uncertainty and to freely collaborate in transdisciplinary teams to solve a systemic problem towards meeting the SDGs, which is considered a desirable skill for the future (e.g. Laherto & Rasa, 2022; Levrini et al., 2021; McWilliam et al., 2008; Rasa et al., 2022). This is necessarily due to the methodological set-up of I2P, in which diverse student teams must independently establish their own collaboration and information-sharing dynamics and organize their approach to the challenge. However, while learning to navigate conversation in such an environment, students requested more grounding via prototyping and hands on activities. It was also apparent in the feedback that the same themes in learnings were emergent in the "coming back to Earth," which suggests the same skills were applied in both tasks. Feedback concerning "coming back to Earth", in comparison to the previous part, no longer emphasized the challenges and difficulty, and as such, the developed skills of openness and dealing with uncertainty were easier put to practice towards meeting current complex problems. While the transfer of learnings was not directly included as a query in the data collection, it can be inferred from the results that students did not feel a connection between the futures task and present problem task, but still reported use and learning of similar skills in both, with the results indicating a lesser challenge to use these skills to combat present challenge-based tasks after learning them in the futures oriented half of I2P. I.e., the learned skills of the futures education were implementable in the present. Our findings provide directions to explore in terms of futures education impact on present problem-solving skills, by introducing a methodological structure, in which parallels between skills in futures tasks and present tasks can be explored. Indeed, I2P presents a program structure and first results than can be developed to bridging the noted gap in research of a lack in impact assessment beyond internal states of participants of futures education.

In terms of research on futures education, our findings reveal emergent insights that have been barely explored in research, such as the connection between openness to alternatives and transdisciplinary discussions. While I2P participants always come from diverse backgrounds, the clearest differentiating factor for the I2P task is different disciplinary backgrounds. Indeed, people from different backgrounds have a different orientation towards futures. This bringing together different backgrounds and ways of relating to the future is a key way increase openness to alternatives in the I2P context. This finds support in the knowledge that people already bring in different ways to project onto futures: It is

with the concept of “futurelessness” that Tutton (2023) shows how different social groups are more or less well positioned to perform their vision of the future. In anthropology and sociology, for example, Afrofuturism is an intellectual movement emerging from afro-american people, that focus on recovering from past wounds—mostly from slavery—in order to produce a desirable future for the community (Guitard, 2023). Some of these scenarios rely heavily on a fictional past to imagine a better future, while Africanfuturism focuses primarily less on “what could have been” and more on “what it could be.” Inherently postcolonial, in contrast to other studied utopias demanding radical socio-economic change, black visions of the future constitute a “*looping movement between past and future*” (Davidson, 2023, p.838). Another example that ultimately shows how different visions of one’s future can impact a civilization is sinofuturism. China is just an example of how a nation or culture, after being colonized, has been victim to what is called “cotemporality denial” (Fabian, 2006): First fixed in an eternal past in museums and then, with China and Japan’s economical growth in 1980–1990, forced in another imaginary representation, best exemplified by Ridley Scott’s movie *Blade Runner*, where Orientalism is everywhere in the decor, but no Asian person is actively given a role in the story. In this sense, techno-sinofuturism is merely a westerner’s vision of what Chinese future (or present) should look like. Other types of sinofuturism have also emerged, namely, a post-orientalist one, where indigenous Chinese representations are opposed to western’s ones (Gaffric, 2023). Indeed, echoing Appadurai (2013b), some groups are not powerful enough to shape the future. We can, therefore, speak of a hegemonic vision of the future, which, since the end of the 1980s, has been that of capitalism seen as part of the natural order of things, at least for European societies as studied by Tutton (2023). While the differences between students’ backgrounds vary, they can offer a way to study and leverage how different futures views can increase openness to alternative futures over hegemonic ones in educational methodologies, as thus, contribute to research fields in, for example, anthropology and sociology.

The insights of this research provide information and inspiration for practitioners to develop new types of courses or to incorporate elements of I2P to foster the discovered learning outcomes. Also, the findings support knowledge that sci-fi and narratives are good for supporting engagement with the learning task, and that prototyping is experienced as relevant to grounding otherwise abstract conversations. While prototyping can encourage action based on the futures scenarios (Wathelet, 2020; Wilf, 2019), it is also important to steer away from prototyping that is too product- and technology-oriented, limiting inevitably the potential of freedom in imagining futures (Schultz, 2018; Wathelet, 2020). The results bring forwards also insights which do not directly support previous research findings, such as issues around third party influence and the time requirement for courses like this: It is important to align all potential participating parties on the goals before the program so that the noted pushes for techno-optimism, for instance, can be addressed. In addition, a clear suggestion is to take as much time as possible for such an immersive learning program, as time pressures were considered to result in superficial engagement with the tasks. A way posited by facilitators to mitigate this issue is to allow students more time to get more comfortable with the challenge, which could require fostering, for example, psychological safety in future programs (Bransby et al., 2024; Edmondson, 1999).

5.1. Limitations

As with any research, there are limitations. We acknowledge that the early stage of this explorative work enforces its own limits to the rigor and generalizability of this research. Specifically, three pilot courses were studied and data was collected as reflection session notes right at the ends of programs. First, student desirability in answers can impact results in a way that students write feedback that is not entirely honest, or that students wish to please teachers/facilitators with their reflections. Second, as the feedback is collected right at the end, it reflects the immediate feelings after the program, which is a point at which the students are usually tired, and thus, reflections could change after a certain time period post-program. In addition, some students continued the “back to Earth” part of the courses back at their home universities, and thus, reflections from these students reflect necessarily a different experience. Indeed, while reflections were gathered after each part, there is also a risk the feedback at the end of the second half is interpreted as students and teachers as reflections to give for the entirety of the

program. However, the findings as an outcome of both self-reported reflections and the ethnography data can be considered more rigid due to the mixed-methods confirmations of specific findings.

5.2. Future research

Future research ought to explore also other data formats to study the impacts of courses like I2P that focus on self-directed collaborative open work with systemic challenges in a futures setting and applying the learned skills to similar challenges outside of a futures context. For instance, further thematic analysis of student deliverables or in depth interview data, especially a while after the completion of the program, would be beneficial to capturing details of experiences and learning reflections. This way, the immediate feelings at the completion of the program can be complemented with “more settled” reflections. Indeed, Jähnichen et al. (2022) investigated the relationship between prospection and goal setting in students and found that goals that were set after a prospection exercise were perceived to be more useful to pursue, as well as better aligned with their identity and values, but the effect was largest when looking at long-term goals. These findings underline the potential benefits for students in participating in prospection exercises, such as I2P. Further, this calls for the need to investigate differences between long- as well as short-term effects of participating in future-directed thought experiments like I2P.

In addition, findings could have further validation from purely quantitative studies, such as pre-post surveys with appropriate measurement instruments. For example, the futures consciousness scale (Ahvenharju, 2022) could be used for futures insights, but also surveys on attitudes and felt capabilities in problem-solving and transdisciplinary work. In addition, research ought to be done on more student groups to have a larger picture of impacts, but also for an experimental set-up, it would be especially beneficial to study the differences between student groups doing the second, “coming back to Earth,” part of I2P, others having done the first part and others not, to explore differences and gain insights into how exactly does the futures exercise impact the solving of current challenges. In a similar vein, the effects of I2P being situated at CERN ought to be explored with I2P being run at a different location, since it is argued that our prospectations of futures are spatially situated and materially conditioned (Strydom, 2024). Other ways and narratives could also be explored for “starting anew” instead of a new planet, and instead, opting, for example, for an alternate dimension narrative.

6. Conclusions

Futures education can develop students’ skills in dealing with complex and systemic problems, dealing with uncertainty, working in transdisciplinary groups, and increase their openness to alternatives. These are considered important skills to deal with challenges societies are facing now and even more so in the future, and the new educational methodology, IdeaSquare Planet, was found to be able to deliver many of such aspirational learning experiences for participants. Transformative change now requires that people see different possibilities to exist in the future.

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